
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) achieve strong video understanding, but their empirical efficiency is constrained by the large visual-token sequences required for long videos. Existing token-reduction methods improve this trade-off mainly by pruning low-importance tokens or merging similar ones, yet their comparison point is incomplete: they emphasize redundancy while giving less explicit treatment to the temporal changes and turning points that often determine benchmark answers, and they do not jointly model spatial and temporal token relations. We therefore frame efficient video compression around a complementary principle: similarity identifies redundancy, whereas difference preserves key events. Following this principle, we introduce ST-SimDiff, a training-free framework that first constructs a spatio-temporal graph over visual tokens and then applies two parallel selection paths. Similarity-based representative token selection uses community structure to retain central tokens for stable content, while temporal difference-based event token selection marks sharp content changes so that dynamic transitions remain available to the MLLM. The resulting subset preserves static context and event-specific evidence under a small token budget. Across long-form video understanding benchmarks, ST-SimDiff consistently improves over state-of-the-art token-reduction baselines while also reducing inference time and GPU memory, demonstrating a favorable empirical balance between accuracy and computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

Large Language Models (LLMs) have accelerated the development of Large Vision-Language Models (LVLMs), whose image and video understanding results now support a wide range of multimodal applications (Zhang et al., 2024c; Liu et al., 2024c). In video settings, a common LVLM pipeline samples frames from a video and converts each frame into hundreds or thousands of visual tokens before passing them to the LLM (Bai et al., 2025; Li et al., 2024). This design is accurate but expensive: as duration and resolution increase, the visual-token count grows rapidly, and the resulting computation and storage costs become a practical bottleneck for long-video analysis and real-time interaction (Fu et al., 2024b).

Prior efficiency methods provide important baselines for this cost-performance trade-off, and they can be grouped by the quantitative signal used to remove tokens. Importance-based pruning exploits the observation from FastV (Chen et al., 2024) that attention distributions contain redundancy across LVLM layers, so tokens with lower attention scores in deeper layers can be pruned to reduce computation. Similarity-based merging or selection instead measures redundancy directly among visual tokens, either across adjacent frames or within a single frame. For example, FrameFusion (Fu et al., 2024b) merges similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) selects dominant tokens at the vision-encoder level. These comparisons show that token reduction is effective, but they also suggest that the choice of signal—importance, similarity, or change—determines which evidence remains available to the model.

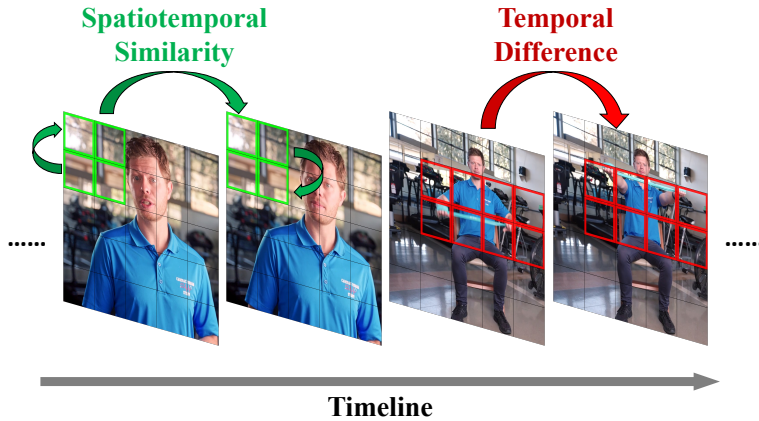


Figure 1: Motivation of ST-SimDiff. The method separates two measurable video patterns that must be handled together for efficient video understanding: spatiotemporal similarity (left), which exposes redundant spatial and temporal content that can be compressed, and temporal difference (right), which identifies action-changing events that should remain in the retained token set.

Despite these gains, existing token-reduction baselines leave two empirical factors under-modeled. First, many approaches compare tokens only within a frame or at the same spatial location across frames, rather than evaluating spatial and temporal correlations jointly. This restricted comparison can miss dynamic events whose evidence is distributed across both dimensions. Second, methods centered on commonality—high importance or high similarity—can underrepresent differences, although video narratives and benchmark questions often depend on abrupt actions or turning points. A similarity-only compression strategy may therefore retain stable background evidence while smoothing over the short-lived changes that distinguish the correct answer.

We address this gap with the following framing: similarity should quantify redundant evidence, while difference should quantify event evidence. An effective video-token compression method should therefore preserve two patterns at the same time: a compact representation of stable content and explicit coverage of key temporal changes. We propose **ST-SimDiff**, a dual token-selection framework built on a spatio-temporal graph. In **similarity-based representative token selection**, all visual tokens are treated as graph nodes, and feature similarity in the spatio-temporal domain defines their connections; community detection then groups persistent visual elements, from which high-centrality tokens are selected as representatives. In **difference-based event token selection**, temporal graph edges are examined for sharp similarity drops between adjacent frames, and tokens at these turning points are retained as event evidence. The final token set merges these representative and event tokens before input to the LLM, making the comparison between stable and dynamic evidence explicit. Our contributions can be summarized as follows:

- We introduce a token-compression perspective that treats similarity and difference as complementary quantitative signals for LVLM efficiency.
- We design a spatio-temporal graph framework that jointly models video-token relationships and supports a dual selection strategy for representative tokens and pivotal event tokens.
- We conduct extensive benchmark comparisons showing that ST-SimDiff substantially reduces the visual-token budget while maintaining or improving video-understanding performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid progress of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has produced a strong generation of Large Vision Language Models (LVLMs)

for multimodal understanding (Liu et al., 2024b; Bai et al., 2023). These models typically encode images or sampled video frames into visual tokens with a pre-trained vision encoder and then feed the tokens into an LLM through alignment modules such as MLPs (Liu et al., 2024a;b) or Q-Formers (Li et al., 2023; Dai et al., 2023). This architecture has delivered strong empirical performance across multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), including increasingly long and high-resolution video understanding benchmarks (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

The same architecture also creates a measurable efficiency challenge. Compared with static images, videos contribute many more visual tokens because the model must process a sequence of frames, and the resulting token count directly increases training and inference cost. As a result, the practical value of LVLMs on long videos depends not only on benchmark accuracy but also on how well the method controls visual-token volume and its downstream computational overhead.

2.2 VISUAL TOKEN COMPRESSION

Visual token compression has become a central approach for improving the accuracy-efficiency trade-off of Large Vision Language Models (LVLMs). Existing methods mainly differ in the quantitative criterion they use for comparison: some rank tokens by importance, while others measure redundancy through token similarity.

Early approaches focused on importance-based pruning, where attention scores or related feature statistics identify tokens that contribute less to model predictions. Representative methods include FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). More recent work (Shang et al., 2024; Ye et al., 2025) adds similarity as an explicit redundancy signal, recognizing that tokens can be individually important yet still duplicate one another. FrameFusion (Fu et al., 2024b), for instance, combines similarity-based merging with importance-based pruning; VisionZip (Yang et al., 2025) keeps dominant tokens while merging contextual information; and VisPruner (Zhang et al., 2025b) uses visual cues to remove redundant items.

Although these baselines make clear empirical progress in reducing redundant visual tokens, they do not fully exploit global spatio-temporal similarity in video. Redundancy appears both within frames and across time, but similarity-driven methods such as FrameFusion primarily compare temporally adjacent content, leaving richer spatial-temporal relationships less directly modeled. This limitation becomes especially consequential under aggressive token budgets, where small differences in the retained evidence can have a large effect on downstream video-understanding accuracy.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video into a full visual-token sequence $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$ of length N . This full-token setting is the accuracy-oriented reference point but is computationally expensive because self-attention scales quadratically with token count, i.e., $O(N^2)$. We therefore define the task as learning or designing a token selection function $f(\cdot)$ that maps T_{full} and a compression ratio r to a smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The target subset should keep downstream LVLM performance as high as possible while satisfying the budget constraint $|T_{\text{sub}}| \approx r \cdot N$. The central design question is which measurable token evidence should be retained so that the reduced subset T_{sub} preserves the information most useful for evaluation.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As shown in Figure 2, ST-SimDiff compares visual tokens through a shared spatio-temporal graph and then selects evidence through two parallel components: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). The two components are designed to cover complementary empirical patterns in video data rather than competing criteria. In SRTS, the method compresses stable and redundant content by detecting communities of highly similar tokens in the graph. These communities correspond to persistent visual evidence such as

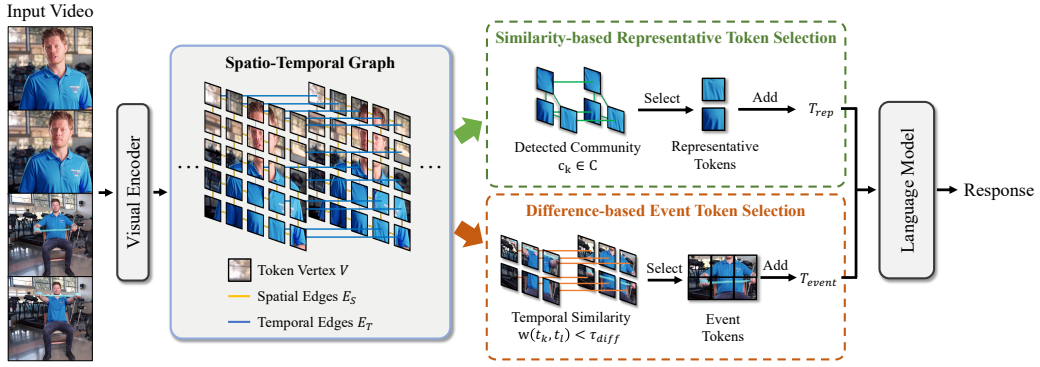


Figure 2: Overview of ST-SimDiff. The framework makes the comparison signals explicit: a spatio-temporal graph first quantifies local token relations, Similarity-based Representative Token Selection retains central tokens from redundant communities, and Difference-based Event Token Selection adds tokens located at strong temporal changes.

static backgrounds or repeated objects, and a small number of high-centrality tokens from each community form the representative set T_{rep} . In DETS, the method measures dynamic evidence by inspecting temporal graph edges for sharp drops in similarity between corresponding tokens in adjacent frames. Tokens around these high-difference transitions are retained as critical event tokens, forming T_{trans} . The final subset $T_{sub} = T_{rep} \cup T_{trans}$ therefore combines compact coverage of stable context with explicit preservation of temporal turning points, yielding a token set that is both budget-aware and information-focused.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

For a given video, the vision encoder produces tokens $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ has a feature vector $\mathbf{x}_k$ and a known spatio-temporal location, represented by its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$. These indices allow token comparisons to be tied directly to the spatial and temporal structure of the video.

We encode these comparisons in a spatio-temporal graph $G = (V, E)$, where the vertex set V contains all tokens in T . The edge set E is decomposed as $E = E_S \cup E_T$, with E_S measuring spatial adjacency and E_T measuring temporal continuity. The spatial edges E_S connect tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens at the same spatial position in adjacent frames, providing the comparisons used to detect continuity and change:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each edge weight $w(v_i, v_j) \in E$ is the cosine similarity between the corresponding token features:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse construction keeps the graph inexpensive while retaining the two comparison axes needed for the method: spatial similarity within frames and temporal similarity or difference across frames.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video tokens contain substantial spatio-temporal redundancy, and this redundancy is measurable through similarity. In a static scene, background tokens remain similar over time; for a moving

object, tokens may shift spatial positions while preserving semantic correlation. Existing methods often evaluate these patterns separately in space or time, whereas our graph places both relationships in a single comparison structure. Dense graph communities therefore provide an empirical signal of repeated content, and selecting representatives from each community compresses redundancy while preserving the central semantics of stable visual evidence.

To identify strongly correlated token groups, we first threshold the graph G . A similarity threshold τ_{sim} keeps only edges whose weights exceed the threshold, producing $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. Next, we apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) on G' to obtain token clusters $C = \{c_1, c_2, \dots, c_m\}$. Within each community $c_k \in C$, we rank tokens by centrality so that the retained tokens are those most representative of the community. The centrality score $S_c(t_a)$ of token $t_a \in c_k$ is the average similarity between t_a and the other tokens in that community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

The intra-community retention rate is set to the global compression ratio r , so each community contributes the top $\lceil |c_k| \cdot r \rceil$ tokens by centrality. This proportional rule applies uniformly across large redundant communities and single-node communities, where the unique token is retained. The representative token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. This selection rule makes the compression margin interpretable: repeated visual evidence is reduced to its most central tokens, while isolated evidence is not discarded simply because it lacks redundant neighbors.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

Similarity identifies what remains stable in a video, whereas difference identifies what changes. This distinction is important for video understanding because a method that preserves only the most common or similar evidence can describe “what is” while missing “what happened.” Events such as object appearance, action onset, or scene transition create localized feature changes, and these changes often determine answers to temporal questions such as “when” and “why.” ST-SimDiff therefore treats high temporal difference as a positive selection signal rather than as noise.

We measure difference along the temporal dimension because adjacent-frame comparisons directly expose event boundaries. Specifically, we analyze temporal edges (E_T) in the spatio-temporal graph. For a temporal edge $(v_k, v_l) \in E_T$ connecting temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the temporal difference exceeds this threshold, equivalently when $w(t_k, t_l) < \tau_{\text{diff}}$, the subsequent token t_l is treated as a critical event token and retained. Formally, the event token set T_{event} is:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This rule preserves tokens at measurable temporal discontinuities, ensuring that aggressive compression does not remove the dynamic evidence that distinguishes one event state from the next.

4.5 OVERALL REDUCTION PROCESS

The full reduction process first computes T_{rep} and T_{event} in parallel and then forms the candidate set $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$. To match the target budget $N_{\text{target}} = \lceil r \cdot N \rceil$, we add a final pruning step. If $|T_{\text{candidate}}|$ exceeds N_{target} , the method removes the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), token importance is determined by attention scores in the shallow layers of the LLM. This final step keeps the retained set aligned with the requested computational budget while preserving the candidate evidence selected by similarity and difference.

Overall, ST-SimDiff combines graph-based selection with attention-based budget adjustment. The graph stage uses the intrinsic structure of the video to preserve representative similarity evidence and event-specific difference evidence, while the attention stage calibrates the final count to the target ratio. This design makes the efficiency-performance trade-off explicit and adaptable across token budgets.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational cost of ST-SimDiff is concentrated in three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS). Each stage is designed to keep the overhead small relative to the LVLM inference cost that token reduction is meant to reduce.

Graph construction computes cosine similarity only between spatially and temporally adjacent neighbors for each of the N visual tokens. Because each token has a small constant number of such neighbors, the graph can be built with one pass over the token sequence, yielding complexity $O(Nd)$, where d is the token feature dimension.

After graph construction, SRTS identifies token clusters. As detailed in our implementation (Section 5.2), we use a connected components algorithm with near-linear complexity $O(N + |E|)$. Since $|E|$ is at most $3N$, this term simplifies to $O(N)$. The within-community filtering step could be expensive under naive all-pairs similarity, with complexity $O(|C| \cdot |c_k|^2 \cdot d)$ for community c_k . To keep this term controlled, communities larger than a predefined threshold (e.g., $|c_k| > \sqrt{N}$) are partitioned. Although graph clustering methods such as Louvain and Leiden can provide richer community definitions, their complexity approaches $O(N \log N)$, so the connected-components choice favors lower overhead and a clearer efficiency margin.

DETS scans temporal graph edges once and compares each temporal similarity with the threshold used to identify significant changes. This event-selection pass has linear complexity $O(Nd)$.

Taken together, the dominant operations in ST-SimDiff are linear in the number of tokens, giving overall complexity $O(Nd)$. This is substantially below the $O(N^2d)$ self-attention cost of the Large Language Model, so the method can reduce token volume while adding only a small preprocessing cost relative to the main inference computation.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines We evaluate ST-SimDiff on two base LVLMs, LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), and compare it with state-of-the-art video token compression methods under matched token budgets. The comparison covers **importance-based reduction methods**, including FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods**, including VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). We also report the original uncompressed model (Vanilla) as the full-token upper-bound reference and random sampling with the same retained-token count as a lower-bound reference.

Benchmarks To test whether the observed gains hold across evaluation conditions, we use three long-video benchmarks: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME reports foundational video understanding across Short, Medium, and Long video categories, allowing comparisons by duration. LongVideoBench evaluates detail retrieval and reasoning in long videos through a “referring reasoning” task, with video lengths of 15s, 60s, 600s, and 3600s. EgoSchema focuses on long-term first-person question answering and measures whether a model can recover character intentions and causal action chains.

5.2 IMPLEMENTATION DETAILS

We run all inference on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04 and use the *lmms-eval* (Zhang et al., 2024a) library for evaluation. LLaVA-Video follows its original 64-frame

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Long-form video understanding comparison on LLaVA-Video-7B with 64 input frames (%). Results are grouped by token retain ratio, and the best method in each benchmark column is emphasized in **bold**.

input setting, and NVILA is also evaluated with 64 input frames so that model comparisons use the same video-frame condition. Unless otherwise stated, the similarity threshold is $\tau_{\text{sim}} = 0.8$ and the difference threshold is $\tau_{\text{diff}} = 0.2$. We report the two practical compression settings $r = 30\%$ and $r = 50\%$, and we use connected component finding for community detection to balance runtime overhead with performance gain.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

Table 1 and Table 2 report the main benchmark comparisons. The evaluation varies four factors explicitly: base model (LLaVA-Video-7B or NVILA-8B), benchmark (VideoMME, LongVideoBench, or EgoSchema), metric column within each benchmark, and token retain ratio ($r = 30\%$ or $r = 50\%$). This structure lets us compare ST-SimDiff against importance-based and hybrid baselines under the same retained-token budget.

(1) ST-SimDiff is the strongest compressed method across the reported configurations. On LLaVA-Video-7B, it obtains the best overall scores at both $r = 30\%$ and $r = 50\%$ on VideoMME, LongVideoBench, and EgoSchema. At $r = 30\%$, for example, ST-SimDiff reaches 63.2 on VideoMME overall, 57.5 on LongVideoBench overall, and 56.0 on EgoSchema, improving over the strongest competing compressed baseline in each corresponding column. At $r = 50\%$, the method reaches 63.8, 57.9, and 57.3 on the same three overall metrics, matching or exceeding the full-token LLaVA-Video reference on VideoMME and EgoSchema while retaining only half of the tokens.

(2) The baseline pattern is consistent with the signals each method uses. Among importance-based methods such as FastV and FasterVLM, FasterVLM is usually the stronger comparator, but importance scoring can keep repetitive tokens that are individually salient yet temporally redundant. Hybrid methods such as PruMerge, VisionZip, and FrameFusion are more competitive because they incorporate redundancy reduction, with FrameFusion often serving as the strongest non-ST-SimDiff

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Long-form video understanding comparison on NVILA-Video-8B with 64 input frames (%). The table contrasts ST-SimDiff with importance-based and hybrid compression baselines at matched token retain ratios, with the best score in each column shown in **bold**.

baseline. However, these methods still primarily optimize for representative commonality; they do not explicitly add a difference-based comparison that protects abrupt events and turning points.

(3) Overall, the comparisons show that modeling redundancy alone is not sufficient for the strongest accuracy-efficiency trade-off. Existing baselines increasingly capture redundant evidence, but they lack a joint spatio-temporal similarity analysis and rarely treat temporal difference as a retained-token criterion. ST-SimDiff’s gains are therefore best understood as the combined effect of these two additions: the graph identifies richer redundant communities, and the difference path preserves key changes that can drive benchmark answers.

5.4 ABLATION STUDY

Table 5 isolates the contribution of each selection signal. The baseline uses only fundamental importance-based pruning; +Sim adds similarity-based selection; and ++Diff adds the proposed temporal difference selection on top of the best similarity setting. We report both $r = 30\%$ and $r = 50\%$ to show how the component margins change as the token budget becomes more or less restrictive.

(1) The +Sim rows show a consistent improvement over the importance-only baseline. At $r = 30\%$, spatial, temporal, and joint spatio-temporal similarity all improve the three reported metrics, and the joint Spa. + Tem. variant is the strongest similarity-only setting, reaching 62.6 on VideoMME, 57.0 on LongVideoBench, and 55.3 on EgoSchema. The same pattern holds at $r = 50\%$, where Spa. + Tem. produces the best or tied-best +Sim scores, supporting the value of joint spatio-temporal redundancy modeling.

(2) Adding ++Diff yields the largest additional margin in the stricter $r = 30\%$ setting: relative to Spa. + Tem., the full method improves VideoMME from 62.6 to 63.2, LongVideoBench from 57.0 to 57.5, and EgoSchema from 55.3 to 56.0. At $r = 50\%$, the gains are smaller but still positive, moving the same metrics from 63.7 to 63.8, 57.8 to 57.9, and 57.2 to 57.3. This budget-dependent

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation results (%). Starting from an importance-pruning baseline, the table isolates the contribution of similarity modeling (+Sim), compares spatial, temporal, and joint spatio-temporal variants, and then reports the additional effect of the proposed difference module (++)Diff) at $r = 30\%$ and $r = 50\%$.

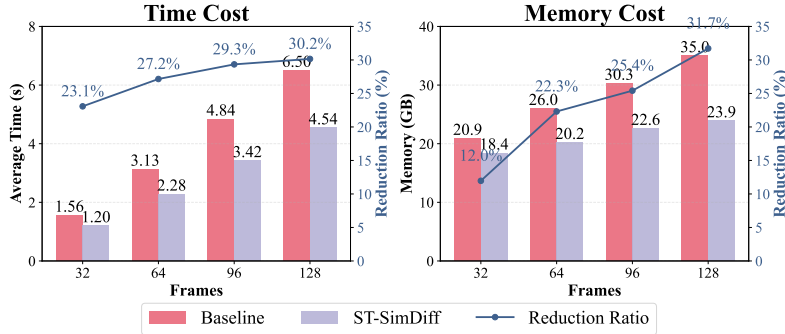


Figure 3: Computational cost comparison between ST-SimDiff and the uncompressed LLaVA-Video baseline under a 30% token budget, measured by average inference time and peak GPU memory usage across 32 to 128 input frames.

pattern supports the intended role of the difference module: it contributes most when compression is aggressive and event tokens are easiest to lose, while still adding stable margins when more tokens are retained.

5.5 COMPUTATIONAL COST ANALYSIS

Figure 3 quantifies the practical efficiency margin of ST-SimDiff under the $r = 30\%$ budget. The comparison uses the uncompressed LLaVA-Video baseline as the reference and reports average inference time and peak GPU memory across input lengths from 32 to 128 frames.

Inference Time ST-SimDiff reduces latency relative to the uncompressed baseline, and the margin grows with video length. The baseline increases from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames, whereas ST-SimDiff reaches 4.54 seconds at 128 frames. The corresponding time-saving rate rises from 23.0% at 32 frames to 30.2% at 128 frames, indicating that token reduction becomes increasingly valuable for longer inputs.

Memory Cost ST-SimDiff also lowers peak GPU memory relative to the uncompressed reference. Baseline memory increases from 20.9 GB at 32 frames to 35.0 GB at 128 frames, while ST-SimDiff uses 23.9 GB at 128 frames. This corresponds to a 31.7% memory saving at the longest evaluated input, which is practically significant for running longer videos on memory-limited hardware.

Taken together, the accuracy tables and cost measurements show that ST-SimDiff improves the empirical trade-off on both axes: it produces leading compressed-model performance while reducing the latency and memory costs that limit long-video LVLM deployment.

6 CONCLUSION

We presented ST-SimDiff, a training-free token selection framework for efficient long-video understanding with Multimodal Large Language Models. The method reframes compression as a quantitative balance between two evidence types: similarity-based representatives for redundant stable content and difference-based event tokens for temporal turning points. Across VideoMME, LongVideoBench, and EgoSchema, and across LLaVA-Video and NVILA backbones, the reported comparisons show that this dual signal yields stronger compressed-model performance than existing importance-based and hybrid baselines under matched token retain ratios. The computational analysis further shows practical savings in inference time and peak GPU memory, especially as input length increases. These results support spatio-temporal similarity and temporal difference as complementary signals for improving the accuracy-efficiency trade-off in video LVLMs.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

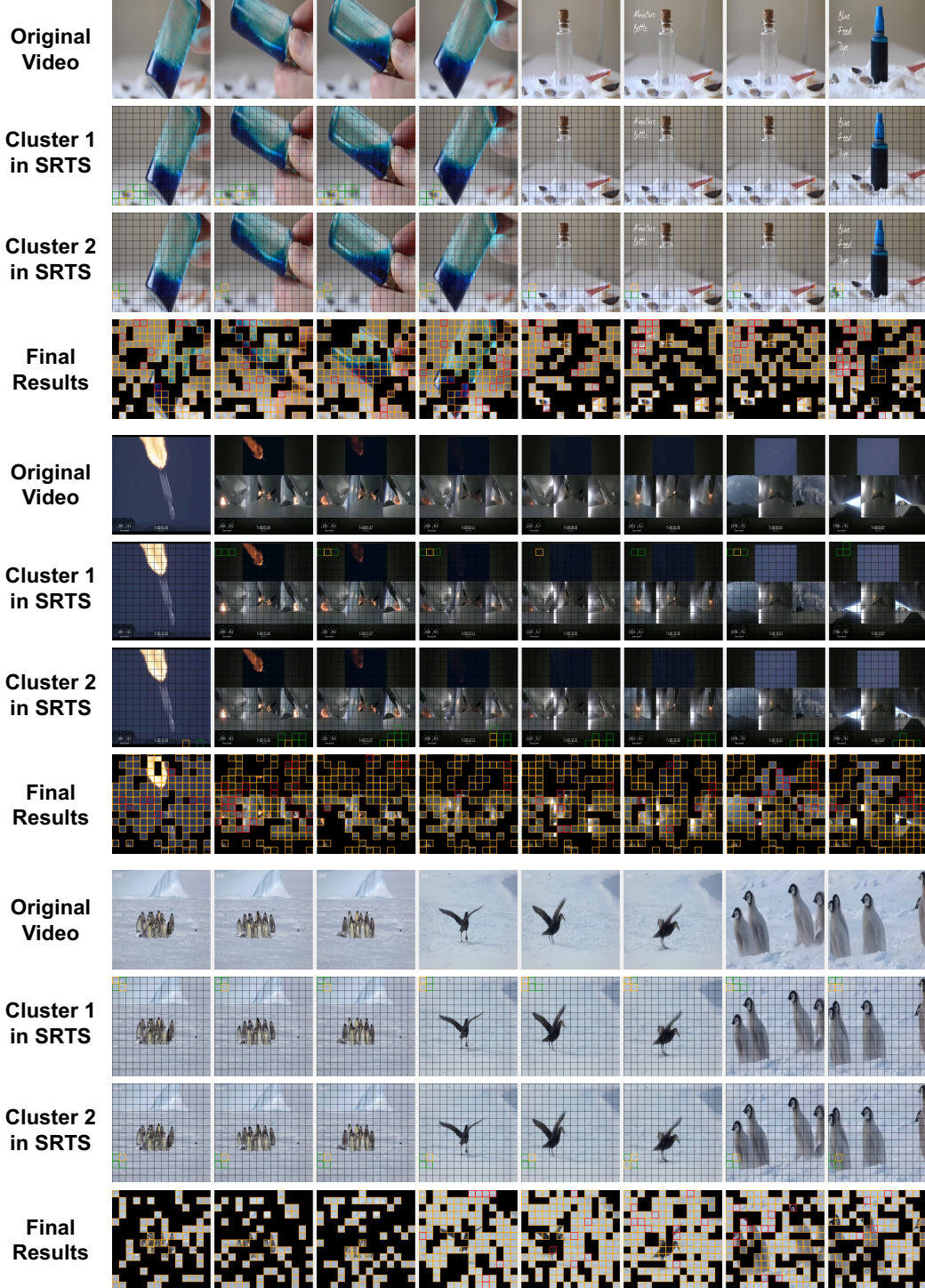


Figure 4: Visualization of ST-SimDiff token selection. The examples separate the two retained-token sources: SRTS communities (Cluster 1 and Cluster 2 grids) identify redundant stable regions, while DETS marks high-difference event tokens; the final overlay contrasts sparse representative tokens in yellow with dense dynamic-event tokens in red.

Figure 4 provides qualitative evidence for how the quantitative selection signals operate before the final token subset is formed. In the dynamic object-manipulation example, the SRTS rows labeled “Cluster 1” and “Cluster 2” show that graph community detection groups spatially and temporally redundant evidence, including static background regions and stationary objects. The sparse yellow boxes in the final result are the retained central representatives, illustrating how stable content can be compressed without keeping every redundant token.

The same visualization also shows the complementary DETS signal. In the “Final Results” row, dense red boxes indicate tokens selected because their temporal difference exceeds the threshold (τ_{diff}). These tokens align with the moving hand and shifting blue liquid, making clear that the difference path preserves fine-grained action evidence that would be easy to lose under similarity-only compression.

The overlay therefore makes the practical comparison visible: yellow representative tokens cover stable context with few samples, while red event tokens concentrate capacity on rapid changes. This pattern supports the empirical framing of ST-SimDiff as a balance between redundancy removal and event preservation, rather than a single-signal pruning strategy.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Long-form video understanding results on Qwen2.5-VL-7B for 64 and 128 input frames (%). Full-token rows provide the upper-bound references, and ST-SimDiff rows report matched token retain ratios of $r = 30\%$ and $r = 50\%$.

To test whether the empirical pattern transfers beyond the main backbones, we additionally evaluate ST-SimDiff on Qwen2.5-VL. The comparison in Table 4 uses VideoMME, LongVideoBench, and EgoSchema with both 64-frame and 128-frame inputs, allowing the token-budget results to be read against full-token Qwen2.5-VL references.

The Qwen2.5-VL results preserve the favorable accuracy-efficiency trade-off. With 64 input frames, ST-SimDiff scores 62.4 on VideoMME overall and 56.8 on LongVideoBench overall at $r = 30\%$; at $r = 50\%$, these scores increase to 62.9 and 58.4. The 50% VideoMME result matches the 64-frame full-token reference of 62.9, while the LongVideoBench result remains close to the 59.2 full-token reference. With 128 input frames, the same table shows strong retained-token performance, including 66.3 on VideoMME overall and 64.5 on EgoSchema at $r = 50\%$.

C ABLATION STUDY RESULTS ON NVILA

We repeat the component ablation on NVILA-Video-8B (Liu et al., 2024c) to check whether the margins observed in the main ablation hold for a different base model. The setup matches the main-paper ablation: the Baseline uses fundamental importance-based pruning, + Sim adds similarity selection, and ++ Diff adds the temporal difference module.

Table 5 shows the same component ordering on NVILA-Video-8B as on LLaVA-Video-7B. At $r = 30\%$, the Spa. + Tem. similarity variant improves the baseline from 59.3 to 59.9 on VideoMME,

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Component ablation on NVILA-Video-8B (%). The table reports the same baseline, +Sim variants, and ++Diff setting as the main ablation, enabling a direct comparison of component margins across token retain ratios.

from 53.8 to 54.6 on LongVideoBench, and from 50.2 to 51.3 on EgoSchema. Adding ++ Diff raises the three metrics further to 60.2, 55.2, and 51.7. At $r = 50\%$, Spa. + Tem. reaches 61.4, 56.0, and 52.3, and the full method improves these to 61.7, 56.5, and 52.5. These consistent positive margins indicate that the similarity and difference modules generalize across base architectures.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

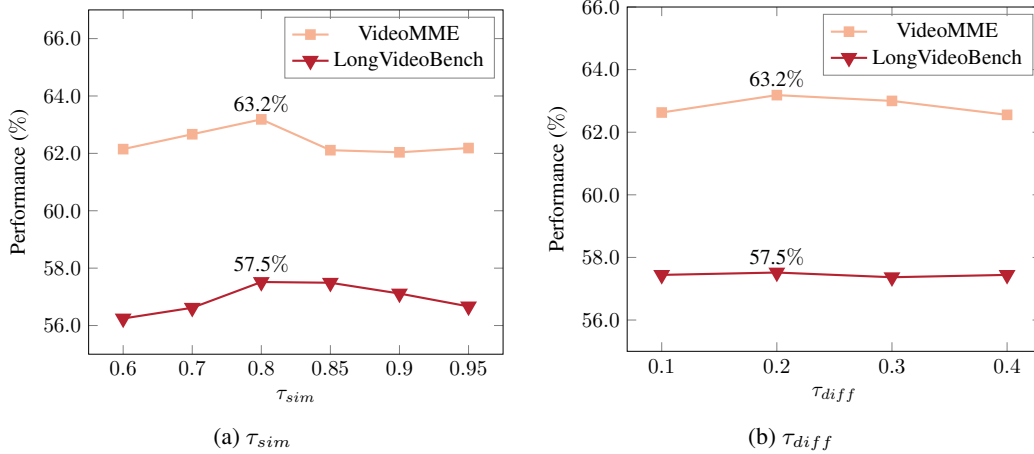


Figure 5: Sensitivity of LLaVA-Video results to τ_{sim} and τ_{diff} on VideoMME and LongVideoBench. The curves compare performance across threshold values while keeping the evaluation benchmarks fixed.

Figure 6 evaluates how the two threshold hyperparameters affect LLaVA-Video performance. Both curves show a stable peak region rather than a brittle single point: $\tau_{sim} = 0.8$ gives the best reported values of 63.2% on VideoMME and 57.5% on LongVideoBench, and $\tau_{diff} = 0.2$ gives the same peak values. Lower τ_{sim} makes community detection less selective, whereas higher τ_{sim} can split semantic clusters; lower τ_{diff} can over-select noisy changes, while higher τ_{diff} can miss event evidence. We therefore use $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ in the other experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

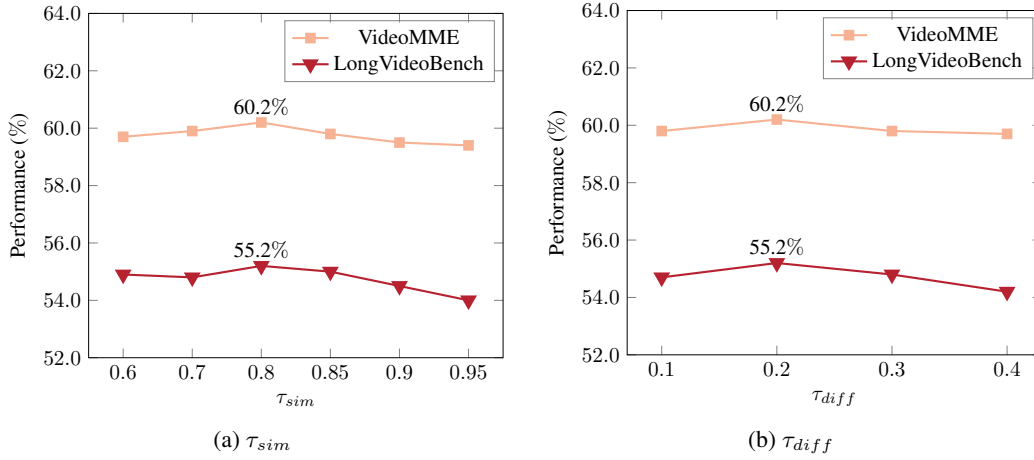


Figure 6: Sensitivity of NVILA-Video-8B results to τ_{sim} and τ_{diff} on VideoMME and LongVideoBench. The plots report how benchmark performance changes as each threshold varies.

We perform the same threshold sensitivity analysis on NVILA-Video-8B to verify that the selected values are not specific to LLaVA-Video. Fig. 6 compares VideoMME and LongVideoBench performance as τ_{sim} and τ_{diff} vary.

Impact of Similarity Threshold τ_{sim} . Across $\tau_{sim} \in [0.6, 0.95]$, NVILA-Video-8B remains relatively stable, with the best scores occurring at $\tau_{sim} = 0.8$. At this threshold, VideoMME reaches 60.2% and LongVideoBench reaches 55.2%. Performance declines modestly when the threshold is lower or higher, indicating that the method benefits from a selective but not overly restrictive similarity criterion.

Impact of Difference Threshold τ_{diff} . Across $\tau_{diff} \in [0.1, 0.4]$, the best NVILA-Video-8B scores occur around $\tau_{diff} = 0.2$, where VideoMME reaches 60.2% and LongVideoBench reaches 55.2%. This pattern matches the intended role of the threshold: values near 0.2 retain salient event changes without over-selecting noisy temporal variation.

Overall, the NVILA sensitivity results confirm that $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ provide strong and consistent benchmark performance across both datasets, reinforcing the same hyperparameter choice used in the main experiments.